

DATA
SET

DS1040

11. Reaction Times A comparison of reaction times (in seconds) for two different stimuli in an experiment produced the following results when applied to a random sample of 16 people:

Stimulus 1	1	3	2	1	2	1	3	2
Stimulus 2	4	2	3	3	1	2	3	3

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

DATA
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DS1041

12. Reaction Times II Refer to Exercise 11. Suppose that the experiment is conducted using eight people as blocks and making a comparison of reaction times within each person; that is, each person is subjected to both stimuli in a random order. The reaction times (in seconds) for the experiment are as follows:

Person	Stimulus 1	Stimulus 2
1	3	4
2	1	2
3	1	3
4	2	1
5	1	2
6	2	3
7	3	3
8	2	3

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

$$\bar{X}_1 = 1.875, S_1^2 = 0.6964, \frac{0.8393}{0.6964} < 3$$

$$\bar{X}_2 = 2.625, S_2^2 = 0.8393$$

$$H_0: \mu_1 - \mu_2 = 0, H_a: \mu_1 - \mu_2 \neq 0$$

$$\alpha = 0.05$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1 + n_2 - 2}$$

$$S_p^2 = 0.76783$$

$$t^* = \frac{1.875 - 2.625}{\sqrt{0.76783 \times \frac{1}{4}}} = -1.7118$$

$$t_{14, 0.025} = 2.145$$

$$RR: \{t \mid |t| > 2.145\}$$

$$|-1.7118| < 2.145$$

$$\Rightarrow \text{no reject } H_0$$